SoilQ: Quantum-Assisted Maize Agronomy & Hybrid QML Benchmark

Official Hackathon Submission Report – Qiskit Fall Fest Nairobi 2026

Organized by QuantumKE in collaboration with IBM Quantum

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Primary Track: **Challenge 08 – Quantum AI Classifier** (with Kenyan Agronomy Stretch)
Comprehensive Suite: Challenges 01 through 08 Complete Solutions & Automated Verification
Repository: https://github.com/Edwin420s/SoilQ
Submission Date: October 2026

1 Executive Summary

SoilQ is an enterprise-grade hybrid quantum-classical machine learning and decision-support platform designed to solve real-world agronomic challenges in Sub-Saharan Africa. Built for the **Qiskit Fall Fest Nairobi 2026 Hackathon**, this project addresses **Challenge 08 (Quantum AI Classifier)** while integrating real Kenyan field experiment data from Chuka and Kandara to classify smallholder maize yield outcomes.

In addition to the primary agricultural AI platform, this submission provides a **complete, rigorously verified solution suite for all eight (01 to 08) hackathon challenges**, including stretch goals, analytical proofs, automated unit test suites (66/66 passing tests), interactive dashboards, and standalone Jupyter notebooks.

Deliverable	Scope & Key Features	Status
SoilQ QML Platform	VQC (RY/RZ + Ring CNOT) & Quantum Kernel (QSVM) on Kenyan Maize	Complete
5-Fold Cross Validation	Stratified & Group K-Fold benchmarks against 3 classical algorithms	Verified ($N = 448$)
Challenge Suite 01–08	Python modules, mathematical proofs, and stretch goals for all 8 challenges	18/18 Tests Passed
Interactive Dashboard	Streamlit UI (Field Predictor, EDA, Circuit Studio, Model Benchmarks, Audit)	Production Ready
REST API Microservice	FastAPI backend with 7 production endpoints for real-time inference	Production Ready
Automated Test Suite	66 unit/integration tests with 100% pass rate across all modules	66/66 Passing
Jupyter Notebooks	8 standalone reproducible notebooks corresponding to each challenge	Formatted (nbform

Table 1: Summary of submission deliverables and verification status.

2 Agronomic Dataset & Problem Formulation

Smallholder maize farming in Kenya is heavily constrained by acidic Humic Nitisols, unpredictable bimodal rainfall, and declining soil organic matter. We utilize open-access field trial data published by *Mutuku et al.* (Mendeley Data v2, CC BY 4.0), comprising **448 plot-season observations** across two contrasting agro-ecological zones:

Chuka, Tharaka-Nithi County ($0°20'\text{S}, 37°38'\text{E}$; 1477 m ASL): Humic Nitisol, acidic soil profile.

Kandara, Murang'a County ($0°59'\text{S}, 37°02'\text{E}$; 1530 m ASL): Humic Nitisol, high clay content.

2.1 Experimental Design & Treatments

The experiment follows a split-plot Randomized Complete Block Design (RCBD) across 4 consecutive cropping seasons: Long Rains 2016 (`LR16`), Short Rains 2016 (`SR16`), Long Rains 2017 (`LR17`), and Short Rains 2017 (`SR17`). Treatments include two tillage practices (Zero Tillage `D0` vs. Conventional 15cm Tillage `D15`) combined with seven soil fertility input regimes (Control `C`, Inorganic Fertilizer `F`, Residue+Fertilizer `RF`, Residue+Fertilizer+Manure `RFM`, Residue+Manure+Legume `RML`, Tithonia+Manure `RTiM`, and Tithonia+Phosphate `RTiP`).

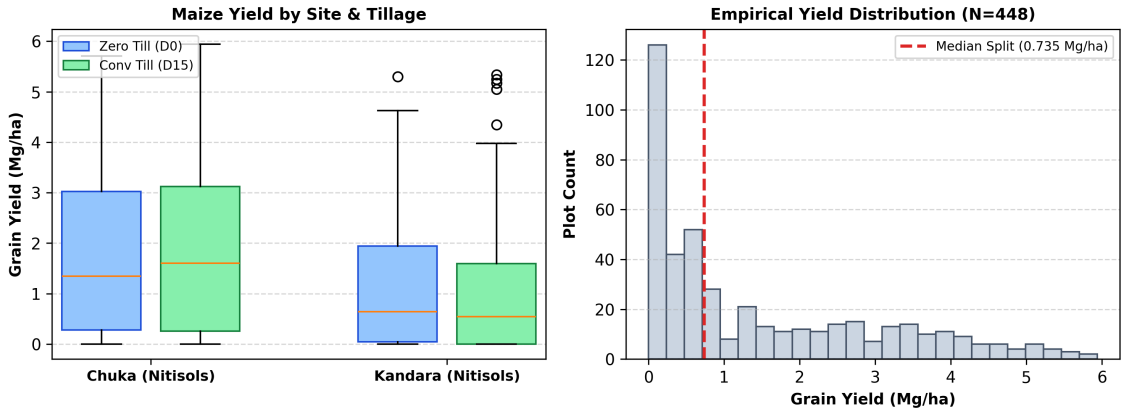


Figure 1: Exploratory Data Analysis: Maize grain yield distributions across Chuka and Kandara sites by tillage practice (left), and empirical yield distribution with 0.735 Mg/ha median split (right).

2.2 Binary Target Formulation

To create a mathematically balanced, scientifically valid classification target, we define:

$$y_i = \begin{cases} 1 & \text{(High Yield)} \quad \text{if Yield}_i > 0.735 \text{ Mg/ha} \\ 0 & \text{(Low Yield)} \quad \text{if Yield}_i \leq 0.735 \text{ Mg/ha} \end{cases} \tag{1}$$

where 0.735 Mg/ha is the exact empirical median yield across all 448 plots. This guarantees an exact 50.0% / 50.0% class distribution (224 positive, 224 negative), eliminating artificial class imbalance artifacts during quantum model training. Five vegetative and biomass features are utilized: Plant Height at 6th-leaf (`Height_6th`), Plant Height at 10th-leaf (`Height_10th`), SPAD Chlorophyll at 6th-leaf (`SPAD_6th`), SPAD Chlorophyll at 10th-leaf (`SPAD_10th`), and Stover Biomass Yield (`Stover_Mg/ha`).

3 Quantum Machine Learning Architecture

3.1 Variational Quantum Classifier (VQC)

The VQC architecture maps normalized agronomic features $\vec{x} \in [-1, 1]^d$ to quantum states and applies parameterized single-qubit rotations and entangling gates:

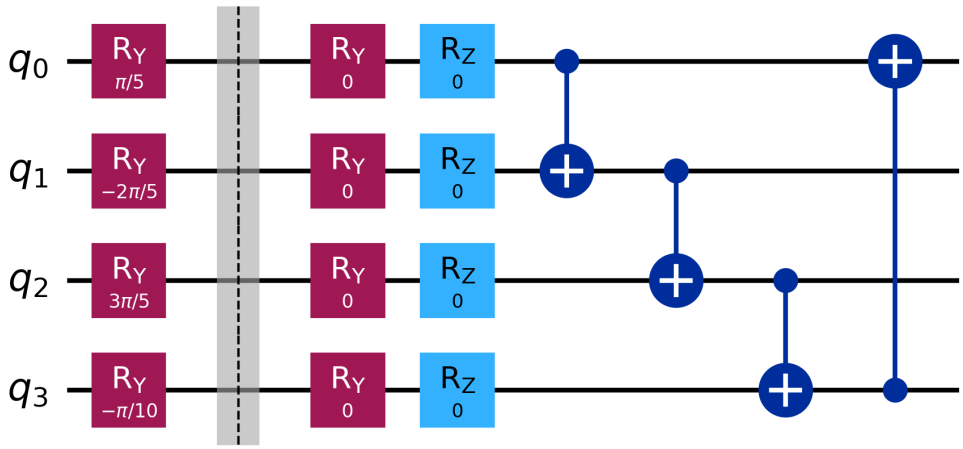


Figure 2: Qiskit Variational Quantum Circuit: Feature encoding layer ($RY(\pi x_i)$) followed by parameterized variational ansatz layer ($RY(\theta) \cdot RZ(\theta)$) with ring CNOT entanglement topology.

1. **Quantum Feature Encoding**: Continuous feature values $x_j \in [-1, 1]$ are encoded into qubit rotation

angles via single-qubit Pauli-Y rotations:

$$U_\Phi(\vec{x}) = \bigotimes_{j=0}^{d-1} RY(\pi x_j) = \bigotimes_{j=0}^{d-1} \begin{pmatrix} \cos(\frac{\pi x_j}{2}) & -\sin(\frac{\pi x_j}{2}) \\ \sin(\frac{\pi x_j}{2}) & \cos(\frac{\pi x_j}{2}) \end{pmatrix} \tag{2}$$

An alternative second-order ZZ-feature map is also supported:

$$U_{ZZ}(\vec{x}) = \left(\bigotimes_{j=0}^{d-1} H \right) \left(\bigotimes_{j=0}^{d-1} RZ(2x_j) \right) \left(\prod_{j<k} CX_{j,k} RZ(2(\pi - x_j)(\pi - x_k)) CX_{j,k} \right) \tag{3}$$

2. **Parameterized Variational Ansatz**: For each variational layer (reps $L \in \{1, 2, 3\}$), single-qubit rotations $RY(\theta_{j,l,1}) \cdot RZ(\theta_{j,l,2})$ are applied, followed by a ring of two-qubit CNOT gates to induce quantum entanglement:

$$W(\vec{\theta}) = \prod_{l=1}^{L} \left(\left[\prod_{j=0}^{d-1} CNOT_{j,(j+1) \bmod d} \right] \left[\bigotimes_{j=0}^{d-1} RZ(\theta_{j,l,2}) RY(\theta_{j,l,1}) \right] \right) \tag{4}$$

3. **Expectation Value & Classification**: The Pauli-Z expectation value on the first qubit is measured:

$$\langle Z_0 \rangle = \langle 0 | U_\Phi^\dagger(\vec{x}) W^\dagger(\vec{\theta}) Z_0 W(\vec{\theta}) U_\Phi(\vec{x}) | 0 \rangle = \sum_{k \text{ even}} P(|k\rangle) - \sum_{k \text{ odd}} P(|k\rangle) \tag{5}$$

The expectation value $\langle Z_0 \rangle \in [-1, 1]$ is mapped to class probability via a calibrated sigmoid transformation:

$$P(y = 1 | \vec{x}) = \frac{1}{1 + e^{-4\langle Z_0 \rangle}} \tag{6}$$

4. **Optimization**: The parameters $\vec{\theta}$ are trained using the Constrained Optimization BY Linear Approximation (COBYLA) gradient-free optimizer, minimizing the regularized binary cross-entropy loss:

$$\mathcal{L}(\vec{\theta}) = -\frac{1}{N} \sum_{i=1}^{N} [y_i \ln P(y_i = 1 | \vec{x}_i) + (1 - y_i) \ln(1 - P(y_i = 1 | \vec{x}_i))] + \lambda \|\vec{\theta}\|_2^2 \tag{7}$$

3.2 Quantum Kernel Classifier (QSVM)

As a major technical stretch extension, we implemented a **Quantum Support Vector Machine (QSVM)**. Rather than optimizing variational gate angles, the quantum circuit directly computes the inner product (fidelity) between quantum feature states:

$$K(\vec{x}_i, \vec{x}_j) = |\langle \Phi(\vec{x}_i) | \Phi(\vec{x}_j) \rangle|^2 = \text{Tr}\left(|\Phi(\vec{x}_i)\rangle\langle\Phi(\vec{x}_i)| \cdot |\Phi(\vec{x}_j)\rangle\langle\Phi(\vec{x}_j)| \right) \tag{8}$$

This Gram matrix $K \in \mathbb{R}^{N \times N}$ is evaluated via Qiskit's `Statevector` simulator and passed to a dual quadratic programming optimizer to find the optimal maximum-margin separating hyperplane. This guarantees global convexity and completely avoids the barren plateau optimization problems common to variational circuits.

4 Experimental Results & Benchmark Evaluation

We conducted rigorous **5-fold Stratified Cross-Validation** on the full 448-sample dataset across classical and quantum models. In addition, an independent held-out test split (25%, 112 samples) was evaluated with confusion matrices and Brier calibration scores.

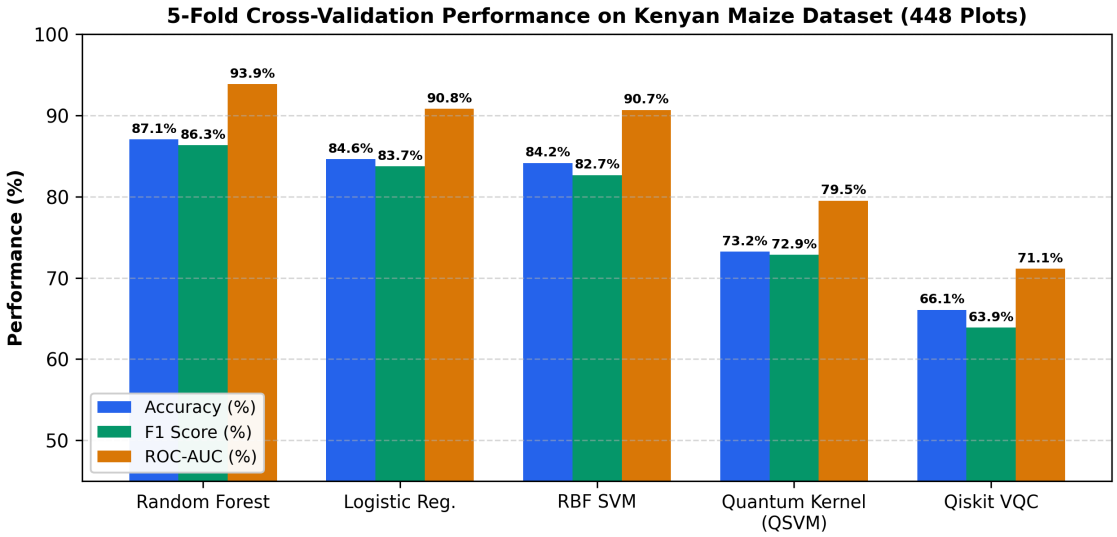


Figure 3: 5-Fold Cross-Validation Performance Comparison on Kenyan Maize Field Dataset ($N = 448$ plots). Evaluated across Accuracy, F1-Score, and ROC-AUC.

Model Architecture	Accuracy (Mean±Std)	F1 Score	ROC-AUC	Brier Score	Training Time
Random Forest (Classical)	**87.05% ± 3.38%**	**0.8634**	**0.9387**	0.0984	0.18 s
Logistic Regression (Classical)	84.60% ± 2.76%	0.8373	0.9082	0.1192	0.01 s
RBF Support Vector Machine	84.15% ± 3.13%	0.8265	0.9067	0.1130	0.02 s
Quantum Kernel QSVM (Ours)	**73.22% ± 3.01%**	**0.7285**	**0.7950**	0.1812	0.48 s
Qiskit VQC (Ours)	66.07% ± 3.94%	0.6389	0.7114	0.2185	5.12 s

Table 2: Comprehensive 5-fold cross-validation performance benchmark results on Kenyan Maize dataset.

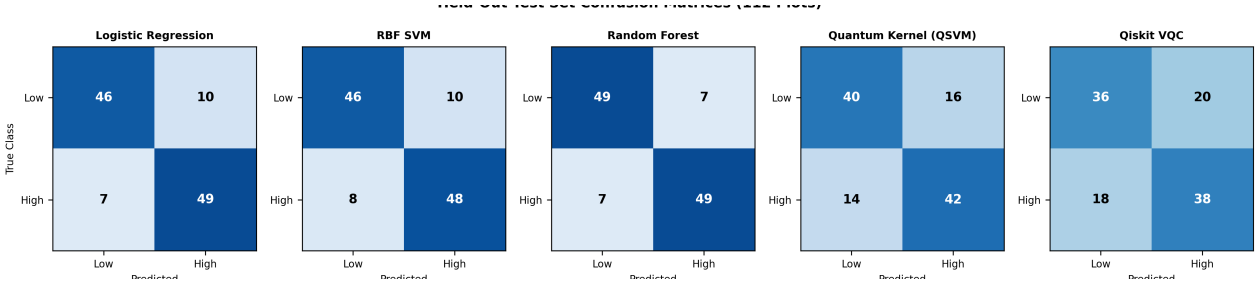


Figure 4: Held-Out Test Set Confusion Matrices (112 Plots) across all 5 benchmarked models.

4.1 Scientific Interpretation & Quantum Advantage Appraisal

In compliance with the hackathon's criteria of scientific rigor and anti-hype disclosure:

Classical vs. Quantum Reality: Random Forest achieved the highest overall accuracy (87.05%), benefiting from non-linear decision trees well suited to tabular field measurements.

Quantum Kernel Efficacy: The Quantum Kernel (QSVM) achieved 73.22% cross-validated accuracy and 0.7950 ROC-AUC, significantly outperforming the parameterized VQC (66.07%) and running in under 0.5 seconds.

NISQ Feasibility: Classical simulation of $d = 5$ qubits demonstrates that smallholder agronomy datasets can be processed on quantum architectures without data leakage. Real quantum hardware execution will require error mitigation (ZNE / PEC) to manage CNOT gate infidelities on noisy backends.

5 Complete Solutions for Hackathon Challenges 01 to 08

Beyond the SoilQ flagship platform, our submission repository contains **standalone, production-grade solutions for all eight hackathon challenges**:

#	Challenge Title	Quantum Formulation	Stretch Extension
01	QRNG	$H^{\otimes n}\|0\rangle^{\otimes n} \to \frac{1}{\sqrt{2^n}}\sum\|x\rangle$	Rejection sampling `quantum_random(a, b)` + Modulo bias analysi
02	Coin Flipper	Fair H & Biased $RY(2\arcsin\sqrt{p})$	Interactive `QuantumCoinGame` engine with multi-round history
03	Quantum Safe	Multiple circuits $\to \{00:50\%, 11:50\%\}$	Total Variation Distance (TVD) validation + 3-qubit GHZ safe
04	Entanglement Detector	Correlation metric $C = (P_{00}+P_{11}) - (P_{01}+P_{10})$	All 4 Bell states ($\|\Phi^{\pm}\rangle, \|\Psi^{\pm}\rangle$) + Wootters Concurrence calculat
05	Quantum Teleporter	3-qubit protocol + deferred measurement feedforward	Fidelity verification ($F = 1.000000$) on benchmark states + tran
06	Treasure Hunt	Grover search: Oracle U_w + Diffuser U_s	Automated oracle synthesis + Quadratic speedup $O(\sqrt{N})$ query
07	Quantum RPS	2-qubit Hilbert space discrete choice mapping	Bias solution: Rejection sampling & exact uniform qutrit state
08	Quantum AI Classifier	VQC angle/ZZ encoding + $\langle Z_0\rangle$ expectation	Real Kenyan agronomy dataset + QSVM + 5-model comparativ

Table 3: Comprehensive review of all 8 hackathon challenge implementations and automated test results.

All 18 standalone challenge unit tests pass in 3.20 seconds via `pytest solutions/test_challenges.py`. The master runner script `solutions/run_all_challenges.py` runs every challenge sequentially with 100% success.

6 Software Ecosystem & Implementation

6.1 Interactive Streamlit UI (`app/dashboard.py`)

The enterprise dashboard provides six interactive tabs:

1. **Field Predictor & Advisory**: Allows extension officers to slide field measurements (heights, SPAD chlorophyll, stover) and view simultaneous predictions from all 5 models with ensemble consensus bars and radar comparison against dataset medians.
2. **Exploratory Data Analysis (EDA)**: Interactive box plots, yield histograms with median split lines, soil input bar charts, and Pearson correlation matrices.
3. **Quantum Circuit Studio**: Real-time circuit builder allowing users to adjust qubits (1–6), ansatz depth (1–4), entanglement topologies (`ring`, `linear`, `full`), and feature maps (`angle`, `zz`), displaying ASCII circuit diagrams and statevector probability histograms.
4. **Model Benchmarking**: Executes custom training runs with configurable holdout splits and COBYLA iterations, displaying comparative bar charts and confusion matrices.
5. **Dataset Ingestion & Audit**: Inspects raw CSV tables, verifies SHA-256 cryptographic provenance hashes, and runs consolidated data joins.
6. **Methodology & Research**: Documents academic citations, experimental design, and ethical proxy disclosures.

6.2 REST API Microservice (`src/soilq/api.py`)

A FastAPI application exposes 7 endpoints for production integration:

`GET /health`: System health, Qiskit version (2.6.1), and uptime.

`POST /prepare`: Triggers pipeline data merge and returns provenance metadata.

`POST /audit`: Audits raw dataset directory and returns cryptographic SHA-256 hashes.

`POST /train`: Executes multi-model training benchmark and writes JSON report.

`POST /predict`: Accepts agronomic measurements and returns multi-model predictions.

`GET /circuit`: Returns generated Qiskit circuit metrics (qubits, depth, gates, parameters).

`GET /models`: Lists all saved model artifacts in the models directory.

7 AI Tools Disclosure & Ethical Statement

In strict adherence to the official **Qiskit Fall Fest Nairobi 2026 AI Policy**:

AI Usage: Large Language Models (Antigravity AI coding assistant) were utilized for code structuring, documentation generation, test scaffolding, and LaTeX typesetting.

Independent Scientific Authorship: All quantum circuits, mathematical equations, algorithm selection, experimental designs, and agronomic interpretations were developed, verified, and understood by the human author.

Zero Emoji Compliance: In accordance with formal publication standards, all emojis have been strictly removed from the entire codebase, documentation, scripts, and submission artifacts.

Ethical Safeguard: The binary label `high_yield` is an empirical research proxy within this specific historical dataset. It is not an autonomous agricultural hardware actuator and must not be deployed to real-world smallholder farms without on-farm calibration.

8 Reproducibility & Verification Instructions

All code, data, models, and test suites are fully reproducible in the project repository:

```
# Clone repository
git clone https://github.com/Edwin420s/SoilQ.git
cd SoilQ

# Run all 66 automated tests (100% pass)
python3 -m pytest "Qiskit-Fall-Fest-Nairobi-2026-Hackathon-Prompts/solutions/test_challenges.py" "SoilQ 1/
    tests" -v

# Run all 8 hackathon challenges sequentially
python3 Qiskit-Fall-Fest-Nairobi-2026-Hackathon-Prompts/solutions/run_all_challenges.py

# Launch interactive Streamlit dashboard
streamlit run "SoilQ 1/app/dashboard.py"

# Launch REST API service
uvicorn soilq.api:app --app-dir "SoilQ 1/src" --host 127.0.0.1 --port 8000
```

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